

Toumadher Barhoumi

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Highlights

- Postdoctoral researcher with a Ph.D. in Systems and Control Engineering and expertise in the modeling, optimization, and control of electrified automotive systems.
- Developed powertrain architecture design methods, energy management strategies, mode shift control strategies, and HILS validation platforms across six sponsored research projects.
- Multiple publications and patents, including an Editors' Choice Award from the ASME Journal of Mechanical Design and patents registered in Korea and the US.
- Principal Investigator on two consecutive KAIST Jang Young-Sil fellowships.

RESEARCH INTERESTS

Electrified propulsion system design and optimization | Optimal, predictive, and learning-based control | Vehicle energy, thermal, and battery management | Connected and automated energy-efficient mobility | Data-driven modeling and optimization | Intelligent transportation systems

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2017 – Feb. 2024

Ph.D. in Systems and Control Engineering (CCS Grad. School of Mobility) Daejeon, Korea

- Advisor: Prof. Dongsuk Kum
- Dissertation: "Design and Control of Parallel/Split Hybrid Vehicle Architectures with Dog Clutches via Virtual Design Space"

Korea Advanced Institute of Science and Technology (KAIST) Sep. 2014 – Feb. 2017

M.S. in Systems and Control Engineering (CCS Grad. School of Mobility) Daejeon, Korea

- Advisor: Prof. Dongsuk Kum
- Thesis: "Automated Physical Realization of Input-, Output-, and Compound-Split Hybrid Configurations"

Korea Advanced Institute of Science and Technology (KAIST) Feb. 2009 – Feb. 2014

B.S. in Aerospace Engineering; Double Major in Electrical Engineering Daejeon, Korea

RESEARCH EXPERIENCE

Mechanical Engineering Research Institute, KAIST Apr. 2024 – Present

Postdoctoral Research Fellow, Automotive Intelligence×Electrification Lab. Daejeon, Korea

- Develop automated topology exploration, component sizing, and charge depleting/sustaining energy management strategies for extended-range electric vehicles (EREVs).
- Build physics-based and semi-empirical dog clutch models to predict engagement behavior and support controller development.

- Optimize dog clutch geometry (e.g., tooth number and taper angle) to improve engagement success and reduce engagement time and impact.

The Cho Chun Shik Graduate School of Mobility, KAIST

Mar. 2017 – Feb. 2024

Graduate Student Research Assistant, Automotive Intelligence×Electrification Lab.

Daejeon, Korea

- Built fuel economy and acceleration performance assessment tools using dynamic programming (DP), equivalent consumption minimization strategy (ECMS), and other optimal control methods.
- Established virtual design space exploration methods for systematic topology optimization and optimal architecture selection of single- and multi-mode power-split hybrid powertrains.
- Developed mode shift controller for a multi-mode power-split hybrid powertrain with dog clutches.
- Built a hardware-in-the-loop simulation (HILS) platform to validate a selected power-split hybrid powertrain configuration and control strategies.
- Developed a schematic design method to systematically enumerate all feasible kinematic diagrams for a given power-split hybrid powertrains with multiple planetary gear sets.

The Cho Chun Shik Graduate School of Mobility, KAIST

Jan. 2014 – Aug. 2014

Research Intern, Vehicle Dynamics and Control Lab.

Daejeon, Korea

- Developed an automated method to depict feasible kinematic diagrams for power-split hybrid vehicles with one planetary gear set.

Korea Aerospace Research Institute

Jun. 2011 – Aug. 2011

Research Intern, Satellite Thermal/Propulsion Department

Daejeon, Korea

- Analyzed candidate semiconductor materials for satellite electronics.

Department of Aerospace Engineering, KAIST

Jun. 2010 – Mar. 2012

Undergraduate Research Program Participant, Aero-Acoustics Lab.

Daejeon, Korea

- Evaluated biodiesel blends for aircraft applications and identified optimal fuel/biofuel blending ratios.

TEACHING AND MENTORING EXPERIENCE

Automotive Intelligence × Electrification Lab., KAIST

Aug. 2024 – Aug. 2026

M.S. Research Mentor

Daejeon, Korea

- Co-mentored an M.S. student on research investigating dog clutch engagement control for a novel multi-mode power-split hybrid vehicle.

The Cho Chun Shik Graduate School of Mobility, KAIST

Spring 2026; Spring 2025

Postdoctoral Teaching Assistant, Modeling and Control of Electric Propulsion Systems (MO50211)

Daejeon, Korea

- Updated existing lecture materials and developed new ones for the revised course syllabus.
- Developed and graded homework assignments and examinations.
- Led laboratory sessions on powertrain modeling and vehicle performance evaluation.

- Led laboratory sessions on powertrain modeling, vehicle performance evaluation, and Hardware-in-the-Loop Simulation (HILS) experiments.
- Held weekly office hours and graded homework assignments and examinations.

PEER-REVIEWED JOURNAL PUBLICATIONS

[J1] **T. Barhoumi** and D. Kum, "Designing a Super-Fast Prius: Systematic Design of Multi-Speed eCVT via Virtual Design Space," *IEEE Transactions on Transportation Electrification*, 2024.
[doi:10.1109/TTE.2023.3255901](https://doi.org/10.1109/TTE.2023.3255901)

[J2] H. Kim, **T. Barhoumi** (co-first author), and D. Kum, "Comprehensive Design Methodology of Compound-Split Hybrid Electric Vehicles: Introduction of the Compound Lever as a Design Tool," *IEEE Access*, 2019. [doi:10.1109/ACCESS.2019.2925146](https://doi.org/10.1109/ACCESS.2019.2925146)

[J3] **T. Barhoumi**, H. Kim, and D. Kum, "Automatic Generation of Design Space Conversion Maps and Its Application for the Design of Compound-Split Hybrid Powertrains," *Journal of Mechanical Design*, 2018.
[doi:10.1115/1.4039451](https://doi.org/10.1115/1.4039451)

[J4] **T. Barhoumi** and D. Kum, "Automatic Enumeration of Feasible Kinematic Diagrams for Split Hybrid Configurations with a Single Planetary Gear," *Journal of Mechanical Design*, 2017.
[doi:10.1115/1.4036583](https://doi.org/10.1115/1.4036583)

MANUSCRIPTS (Under review or in preparation)

[M1] **T. Barhoumi** and D. Kum, "Blending Power-Split and Parallel Systems for Superior Performance: An Exhaustive Search via Virtual Design Space," submitted.

[M2] **T. Barhoumi** and D. Kum, "Beyond Power Transmission: eCVT as a Transitional Mode for a Hybrid System using dog clutches," in preparation.

INTERNATIONAL CONFERENCE PROCEEDINGS (Peer-reviewed, presentations)

[C1] **T. Barhoumi** and D. Kum, "Design and Analysis of a Novel Multi-Speed Output-Split Hybrid Electric Vehicle Powertrain," IEEE ITEC+EATS, 2026. [doi:10.1109/ITECEATS66641.2026.11592898](https://doi.org/10.1109/ITECEATS66641.2026.11592898)

[C2] **T. Barhoumi** and D. Kum, "Systematic Design of Fixed-Gear Modes for Novel Multi-Mode Architectures: A Prius Case Study," *Proceedings of the European Control Conference*, 2022.
[doi:10.23919/ECC55457.2022.9838163](https://doi.org/10.23919/ECC55457.2022.9838163)

[C3] C. Song, J. Hwang, **T. Barhoumi**, and D. Kum, "Complete Design Space Exploration of the Generic Voltec Second-Generation Powertrain via Virtual Levers," AVEC, 2018.

[C4] **T. Barhoumi**, H. Kim, and D. Kum, "Compound-Lever-Based Optimal Configuration Selection of Compound-Split Hybrid Vehicles," SAE Technical Paper 2017-01-1148, 2017. [doi:10.4271/2017-01-1148](https://doi.org/10.4271/2017-01-1148)

[C5] **T. Barhoumi** and D. Kum, "Automated Schematic Design of Power-Split Hybrid Vehicles with a Single Planetary Gear," ASME Dynamic Systems and Control Conference, 2014.
[doi:10.1115/DSCC2014-6086](https://doi.org/10.1115/DSCC2014-6086)

DOMESTIC CONFERENCE PROCEEDINGS

[D1] **T. Barhoumi** and D. Kum, "Bridging the Gap between Power-Split and Parallel Systems for a Superior Performance: Prius Case Study," KSAE Annual Conference, 2024.

[D2] **T. Barhoumi** and D. Kum, “Mode Shift Control of Multi-Mode Power-Split Transmission Using Dog Clutches,” KSAE Annual Conference, 2024.

[D3] **T. Barhoumi** and D. Kum, “Analysis of a Toyota Prius 4th Generation-Based Multi-Mode System Using Dog Clutches,” KSAE Annual Conference, 2021.

[D4] **T. Barhoumi** et al., “Evaluation of the Full Potential of the Generic Voltec Second Generation Powertrain through Exhaustive Virtual Design Space Search,” KSAE Annual Conference, 2018.

[D5] H. Kim, **T. Barhoumi**, and D. Kum, “Optimal Configuration Selection of Two Planetary Gear Compound-Split Hybrid Electric Vehicles,” KSAE Annual Conference, 2017.

PATENTS

[P1] D. Kum and **T. Barhoumi**, “Driving Apparatus for Hybrid Vehicle with Multiple Operating Modes,” Korean Patent Application No. 10-2025-0061716, 2025.

[P2] D. Kum et al., “Multi-Mode Hybrid Vehicle Powertrain Apparatus,” U.S. Patent No. 10,131,220, issued November 20, 2018.

[P3] D. Kum et al., “Hybrid Vehicle Powertrain Apparatus with Multiple Modes,” Korean Patent Registration No. 10-1865862, 2018.

[P4] D. Kum et al., “Driving Apparatus for Hybrid Vehicle Having Multiple Modes,” PCT/KR2016/015492, 2017.

[P5] D. Kum et al., “Hybrid Powertrain Apparatus,” Korean Patent Registration No. 10-1879127, 2017.

[P6] D. Kum et al., “Hybrid Vehicle Powertrain Apparatus with Multiple Modes,” Korean Patent Registration No. 10-1688334, 2016.

RESEARCH GRANTS AND FELLOWSHIPS

Design-Space Exploration and Auto-Sizing Framework for Next-Generation Extended-Range Electric Vehicle (EREV) Powertrains

Dec. 2025 – Nov. 2026

Principal Investigator | Jang Young-Sil Fellowship | KRW 50 million

KAIST

- Enumerating the complete design space of compatible EREV architectures across series, parallel, and power-split CS modes and FWD/RWD/AWD layouts.
- Developing EREV-specific performance metrics (all-electric range, charge-depleting (CD) efficiency, charge-sustaining (CS) fuel economy, gradeability) and scalable component sizing models for engine, battery, and electric machines, coupled into a unified evaluation tool for architecture selection.

Development of World-Class High-Performance Multi-Mode (Plug-In) Hybrid Powertrain Systems Using Dog Clutches

Dec. 2024 – Nov. 2025

Principal Investigator | Jang Young-Sil Fellowship | KRW 50 million

KAIST

- Integrated power-split and parallel-type operating modes to unlock new multi-mode hybrid and plug-in hybrid vehicles with improved fuel economy and acceleration performance.
- Optimized topology, component sizing, and control strategies simultaneously via virtual design space exploration.
- Developed mode shift control strategies addressing torque hole and speed discontinuity.
- Outcomes: 2 journal submissions [M1, M2], 1 international conference paper [C1], 1 patent application [P1].

SELECTED FUNDED RESEARCH PROJECTS

Bullets describe individual contributions.

Development of a Performance Analysis Model for a Dedicated Hybrid Transmission (DHT)

Mar. 2019 – Jul. 2020

Researcher | Funding agency: Hyundai Motor Company

KAIST

- Modeled five representative DHT systems, including a novel transmission.
- Developed fuel economy and acceleration performance assessment tools using DP.
- Derived implementable real-time control and mode shift strategies from DP results.
- Built a forward-looking simulator capturing dog clutch engagement and disengagement in the novel system.

Development of World-Class Multi-Mode Power-Split Hybrid Powertrain Technologies Applicable to HEVs and PHEVs

Mar. 2017 – Feb. 2021

Researcher | Funding agency: National Research Foundation of Korea

KAIST

- Proposed a virtual design space enabling efficient assessment of novel multi-mode hybrid powertrains applicable to both HEVs and PHEVs.
- Enumerated feasible kinematic diagrams for selected multi-mode systems and contributed to hardware-in-the-loop simulation (HILS) implementation of the selected architecture.
- Validated the selected system and its control strategies through HILS.
- Outcomes: 2 conference papers [C2, C3], 3 registered patents [P2, P3, P4].

Exploration and Core-Technology Development of Power-Split (P)HEV Architectures with World-Class Fuel Economy and Acceleration Performance

Jun. 2015 – Mar. 2019

Researcher | Funding agency: Korea Evaluation Institute of Industrial Technology

KAIST

- Introduced the compound lever and automated design space conversion maps to eliminate redundant compound-split architectures.
- Evaluated fuel economy and acceleration performance to select optimal power-split powertrains.
- Developed systematic schematic design methods to generate physical realizations of compound-split powertrain architectures.
- Outcomes: 2 journal articles [J2, J3], 1 conference paper [C4], 2 registered patents [P5, P6].

Development of Novel Power-Split Hybrid Powertrain Architectures for Global Technology Leadership

Sep. 2013 – Oct. 2016

Researcher | Funding agency: National Research Foundation of Korea

KAIST

- Developed automated methods to enumerate all feasible kinematic diagrams for power-split hybrid powertrains with one and two planetary gear sets.
- Outcome: 1 conference paper [C5].

HONORS AND AWARDS

- Jang Young-Sil Fellowship, KAIST — Recipient of two consecutive KRW 50 million research grants, 2024–2026.
- Research Achievement Award, Cho Chun Shik Graduate School of Mobility, KAIST, 2019.
- Editors' Choice Paper Award — Honorable Mention, ASME Journal of Mechanical Design, 2018.
- Boeing Scholarship, 2013.
- Best Lighter Than Air Project Certificate, KAIST MAE292, 2011.
- Award for Outstanding Application of Formal Design Theories, 2010.

PROFESSIONAL SERVICE

Ad hoc Journal Reviewer: *IEEE Access*; *Applied Energy*; *Energy*; *Energy Conversion and Management*; *IEEE/ASME Transactions on Mechatronics* 2024–Present

Memberships: IEEE (Member); IEEE Vehicular Technology Society; IEEE Intelligent Transportation Systems Society; IEEE Young Professionals; IEEE Women in Engineering; Korean Society of Automotive Engineers (KSAE)

TECHNICAL SKILLS

Control and optimization: Dynamic programming (DP), Pontryagin's Minimum Principle (PMP), ECMS, model predictive control (MPC), optimal control, real-time control

Modeling and simulation: Dynamic-system modeling, quasi-static and forward vehicle simulation, powertrain performance analysis, design space exploration, component sizing

Software: MATLAB/Simulink, Python, PyTorch, TensorFlow, C/C++, LaTeX

Hardware and validation: Hardware-in-the-Loop Simulation (HILS), dSPACE tools (MicroAutoBox)

CAD: AutoCAD, SolidWorks

Languages: Arabic (bilingual), French (bilingual), English (fluent), Korean (high intermediate)